

Advanced Deterministic Transformation of Time-Use Activity Sequences into Residential Electricity Load Profiles for Canadian EnergyPlus Simulations

Introduction to Deterministic Time-Use

Transformation in Building Performance Simulation

The accurate modeling of residential electricity demand is a critical imperative for modern energy system design, grid capacity planning, and building performance simulation (BPS). Historically, building simulation engines such as EnergyPlus have relied on static, homogenous fractional schedules to represent internal heat gains and electrical loads. These generalized profiles, often derived from legacy ASHRAE or Department of Energy (DOE) reference buildings, fundamentally fail to capture the extreme temporal volatility, spatial diversity, and appliance-level load geometries inherent to actual human behavior. While stochastic generation models—such as the discrete-time Markov chain frameworks pioneered in the early 2010s—have successfully synthesized high-resolution occupancy and activity states, integrating these models directly into deterministic simulation environments often introduces unwanted computational volatility and decoupling from multi-agent behavioral models. When a multi-agent or predictive behavioral model has already established a rigid, 30-minute sequence of human activities based on General Social Survey (GSS) Time-Use categorizations, the analytical requirement shifts fundamentally from stochastic activity *generation* to deterministic *transformation*. The modeling challenge lies in translating a qualitative, categorical behavioral state (for example, "Household work & maintenance" or "Passive leisure") into an explicit, quantifiable electrical draw (measured in Watts) distributed across specific appliance end-use buckets. Because the simulation architecture already possesses the longitudinal time-use sequence, the application of probabilistic Markov transition matrices or "switch-on dice" is obsolete and counterproductive. Instead, this requires a robust, empirical crosswalk that dictates deterministic appliance-mapping weights, equipment active power ratings, temporal cycle durations, multi-occupant scaling physics, and seasonal corrections perfectly suited to the Canadian climatological context.

This comprehensive report provides an exhaustive, fully quantified framework for mapping predefined 30-minute GSS time-use sequences to high-resolution residential electricity demand profiles. The methodology is heavily anchored in the seminal frameworks established by Richardson, Thomson, Infield & Clifford in their 2010 work, *Domestic electricity use: A high-resolution energy demand model* (Energy & Buildings 42(10), DOI:

10.1016/j.enbuild.2010.05.023)¹, alongside their foundational 2009 domestic lighting model (Energy & Buildings 41(7), DOI: 10.1016/j.enbuild.2009.02.010)³ and 2008 occupancy model

(Energy & Buildings 40(8), DOI: 10.1016/j.enbuild.2008.02.006).⁴ To adapt these principles to modern building simulation, this report integrates the appliance parameterization from McKenna & Thomson's 2016 CREST integrated model (Applied Energy 165, DOI: 10.1016/j.apenergy.2015.12.089)⁵ and the sophisticated behavioral mapping logic presented by Widen & Wackelgard in 2010 (Applied Energy 87(6), DOI: 10.1016/j.apenergy.2009.11.012).⁹ Crucially, to situate this framework within the Canadian built environment, it incorporates baseline validations from Armstrong, Swinton, Ribberink, Beausoleil-Morrison & Millette (2009, J. Building Performance Simulation 2(1)) and the recent synthesis by Osman, Ouf, Azar, and Dong in their 2023 paper, *Stochastic bottom-up load profile generator for Canadian households' electricity demand* (Building & Environment, DOI: 10.1016/j.buildenv.2023.110490).¹⁰ By synthesizing these anchor sources, the following sections establish a rigorous numerical foundation for EnergyPlus implementation, detailing the exact scaling laws required to handle intra-household co-presence, the thermodynamic realities of appliance cycle durations across discrete 30-minute simulation blocks, and actionable directives for validating the synthesized profiles against leading high-resolution Canadian empirical datasets.

The 14-State Categorical Mapping Framework and End-Use Taxonomy

The Canadian General Social Survey (GSS) on Time Use categorizes human behavior into distinct primary activities. For residential load modeling, these qualitative activities must be explicitly mapped to physical end-use buckets. To maintain thermodynamic fidelity within an EnergyPlus zone, the model must differentiate between sensible heat, latent heat, and direct electrical consumption. Therefore, the end-use taxonomy is divided into active, behavior-driven loads and constant, background baseloads.

Defining the End-Use Buckets and Baseload Isolation

The active end-use buckets required for deterministic assignment are: cooking (encompassing the electric range, oven, microwave, and kettle), the dishwasher, the clothes washer, the clothes dryer, television and entertainment systems, computer and home-office equipment, personal-care devices interacting with domestic hot water (DHW), and generalized spatial lighting.

It is a critical mechanical requirement of this framework that the always-on baseload—comprising refrigeration, standalone freezers, and the parasitic standby power of inactive electronics—be handled entirely separately from the activity-driven mapping matrix. Baseload elements operate continuously and independently of human behavioral states.⁸ Assigning a refrigerator's cycling load to a specific human activity like "Eating & drinking" creates artificial and erroneous load spikes. Instead, the total aggregate standby power and refrigeration cycle load (averaging between 84 W and 120 W for a modern Canadian household) must be injected into the EnergyPlus environment as a constant or sinusoidal background vector.¹³ This ensures that the building maintains a realistic base internal heat gain even when the property is flagged as entirely vacant.

Inactive and Exogenous States

Before assigning active load multipliers to the predefined 30-minute sequences, the taxonomy must explicitly quarantine activities that occur entirely outside the residential thermal zone or involve zero active interaction with electrical end-uses. The deterministic model must enforce a zero-multiplier constraint for specific states to prevent energy leakage in the simulation.

The following states map to no active load:

- **State 4 (Purchasing/Shopping):** This activity is exogenous to the building. The active load multiplier is strictly 0.0.
- **State 12 (Community/Volunteer):** This activity is exogenous to the building. The active load multiplier is strictly 0.0.
- **State 13 (Travel):** This activity is exogenous to the building. The active load multiplier is strictly 0.0.
- **State 5 (Sleep):** This is an endogenous but strictly inactive state. While occupants are physically present in the building—thereby generating latent and sensible metabolic heat critical to the EnergyPlus zone heat balance—they trigger zero active appliance loads.¹⁵ Lighting is suppressed, entertainment devices are assumed to be in standby mode (captured by the isolated baseload), and active appliance interaction is halted. The active load multiplier across all end-use buckets for State 5 is 0.0.

The Deterministic Activity-to-End-Use Weight Matrix

For the remaining endogenous active states, the deterministic transformation relies on a comprehensive matrix of relative dependency weights. Because the user is processing pre-determined 30-minute blocks without utilizing stochastic dice-rolling or Markovian transition probabilities, these weights represent the *mathematical expectation value* or *relative load intensity* that a given activity drives a specific end-use.

Within this paradigm, a weight of 1.0 implies that the activity is the sole, primary driver of that appliance class, guaranteeing its utilization (subject to sub-distribution splits). A weight of 0.0 implies no thermodynamic or behavioral relationship. Fractional weights indicate that the activity frequently, but not always, utilizes the appliance, acting as a deterministic smoothing function that distributes the expected energy consumption accurately across the 30-minute block without requiring Monte Carlo sampling.

The following numerical activity-to-end-use weight table synthesizes the appliance mapping logic derived from Canadian household archetype modeling, GSS time-use mapping literature, and the Widen & Wackelgard load distributions.⁹

GSS Code	Activity Description	Cooking	Dish washer	Clothes Washer	Clothes Dryer	TV / Ent.	PC / Office	Personal Care & DHW	Lighting
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1	Work (at home)	0.05	0.0	0.0	0.0	0.0	0.90	0.05	1.0
2	Hous ehold work & maint .	0.10	0.20	0.30	0.20	0.0	0.0	0.20	1.0
3	Care givin g	0.10	0.0	0.0	0.0	0.30	0.0	0.10	1.0
4	Purch asing (Awa y)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5	Sleep (Inact ive)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6	Eatin g & drinki ng	0.85	0.15	0.0	0.0	0.0	0.0	0.0	1.0
7	Perso nal care	0.0	0.0	0.0	0.0	0.0	0.0	0.90	1.0
8	Educ ation (at home)	0.05	0.0	0.0	0.0	0.0	0.85	0.0	1.0

9	Socializing	0.15	0.0	0.0	0.0	0.40	0.0	0.0	1.0
10	Passive leisure	0.0	0.0	0.0	0.0	0.85	0.15	0.0	1.0
11	Active leisure	0.0	0.0	0.0	0.0	0.20	0.0	0.20	1.0
12	Community (Away)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
13	Travel (Away)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
14	Misc. (at home)	0.0	0.0	0.10	0.0	0.10	0.10	0.0	1.0

The allocation of a 1.0 weight to the "Lighting" category across all active endogenous states represents a critical paradigm in deterministic modeling. As established by Richardson et al. (2009), lighting is fundamentally a function of active spatial occupancy combined with environmental irradiance, rather than being strictly tied to the specific mechanics of a given activity.¹ By setting the activity-driven weight to 1.0 for all waking home states, the algorithm acknowledges that the occupant requires spatial illumination. However, this base weight must be subsequently modulated by a solar irradiance override function downstream in the EnergyPlus schedule to ensure lights are not burning at noon on a clear summer day. Furthermore, the nuances in the weight matrix are designed to reflect the complex realities of multi-tasking and secondary load generation. For example, "Work (at home)" heavily drives the

PC/Office bucket (0.90), but retains a fractional 0.05 allocation to Cooking and Personal Care to account for periodic breaks to utilize a kettle, coffee maker, or washroom facilities during a prolonged 30-minute block.¹²

Intra-Activity Sub-Distributions and Probabilistic Disaggregation

A fundamental mathematical limitation of utilizing broad Time-Use Survey categories is their aggregation of highly diverse electrical signatures into a single temporal block. For instance, GSS Category 2 ("Household work & maintenance") represents a 30-minute window that could physically manifest as vacuuming (a high-draw, short-duration inductive motor load), operating a washing machine (a moderate-draw, medium-duration motor and heating load), or hand-washing dishes (a pure Domestic Hot Water pumping and heating load).

Because the EnergyPlus simulation engine requires discrete, appliance-specific equipment scheduling to accurately calculate internal heat gains and zone-specific sensible/latent splits, these broad GSS categories must be fractured into concrete sub-distributions. When translating the deterministic 30-minute activity flag into physical appliance loads, the following fractional probabilities—derived directly from the high-resolution activity clustering analysis by Osman et al. (2023) and supporting European time-use micro-data—must be utilized to split the expected energy value.¹²

Disaggregation of GSS Code 2: Household Work & Maintenance

When the predefined time-use sequence dictates that an occupant is engaged in "Household work," the deterministic energy model must partition this temporal allocation across the dominant cleaning and maintenance end-uses. Based on activity clustering analysis of North American behavioral archetypes¹², the internal fractions are defined as follows:

- **Doing Laundry (Washing Machine):** 0.35
- **Doing Laundry (Clothes Dryer / Ironing):** 0.25
- **Dishwashing (Machine + Hot water interaction):** 0.20
- **Cleaning (Vacuum cleaner + Miscellaneous small motors):** 0.20

In a purely deterministic framework, these fractional probabilities serve as direct power multipliers. Rather than rolling a random number to decide if the washing machine or the vacuum is running during this specific 30-minute block, the base active power of every appliance in this sub-category is multiplied by its respective sub-distribution fraction. This ensures the 30-minute block correctly injects the statistically expected volume of Watts into the thermal zone, conserving energy over the course of the simulation year without inducing artificial stochastic variance.

Disaggregation of GSS Code 6: Eating & Drinking (Meal Preparation)

The "Eating & drinking" classification inherently encompasses both the physical consumption of

food and the intensive process of its preparation. The thermal intensity of food preparation varies drastically depending on the technology deployed, which is often a function of the time of day. Extensive analysis of Canadian residential archetypes differentiates between light, short-duration breakfast loads and intensive, high-thermal-mass dinner loads.¹²

For a generalized, time-independent sub-distribution within the "Eating & drinking" code, the active cooking load should be partitioned across the primary thermal technologies:

- **Electric Range / Oven (High thermal load, extended duration):** 0.45
- **Microwave (Medium load, high frequency short duration):** 0.35
- **Small Appliances (Kettle, Toaster, Coffee Maker):** 0.20

Temporal Override Rule: For advanced EnergyPlus scheduling aimed at capturing peak grid impacts, these static fractions should be modulated based on the time of day to reflect Canadian domestic rhythms. Between the hours of 06:00 and 10:00, the Small Appliance and

Microwave fractions should artificially inflate to 0.50 and 0.40 respectively, reflecting typical morning habits.¹² Conversely, between 16:00 and 20:00, the Range/Oven fraction should

dynamically inflate to 0.70 to represent primary evening meal preparation, effectively shifting the heavy internal heat gains to the early evening hours, aligning with the macroscopic Canadian winter peaking phenomenon.

Disaggregation of GSS Code 10: Passive Leisure

Passive leisure generally involves prolonged engagement with electronic entertainment. While historically dominated by the television, the modern proliferation of secondary screens, mobile devices, and high-draw gaming consoles necessitates a distinct split distribution¹⁵:

- **Television / Home Theater Systems:** 0.65
- **Desktop Computer / Console Gaming (High draw):** 0.20
- **Laptop / Tablet / Mobile Charging (Low draw):** 0.15

Appliance Active Power Ratings, Cycle Durations, and Temporal Discretization

To convert the dimensionless mapping weights and sub-distribution fractions into physical electrical arrays (measured in Watts), precise active power ratings and typical operational cycle durations must be established. The values detailed below prioritize Canadian domestic appliance averages based on Natural Resources Canada (NRCan) Energy Use Equipment standards¹² and the comprehensive high-resolution parameters codified in the CREST v2 integrated domestic demand model developed by McKenna & Thomson.⁶

End-Use /	Typical Active	Standby	Typical Cycle	Thermodyna
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Appliance Type	Power (W)	Power (W)	Duration (Minutes)	mic and EnergyPlus Implementation Notes
Electric Range / Oven	3000 W	2 W	30 - 45	Highly intermittent duty cycle. Thermostatically controlled; the average continuous power over a 30-min block is often 50% of the rated nameplate value.
Microwave	1500 W	3 W	5 - 10	Short duration. If deterministically mapped to a full 30-min block, power must be scaled down by a factor of $\approx$ to preserve the energy balance.
Kettle / Coffee Maker	1200 W	1 W	5 - 15	High peak, rapid decay. Primarily sensible heat gain.
Dishwasher	930 W	2 W	90 - 120	Cycle exceeds

				the 30-min activity block. Requires a mathematical queue function (runs for 3 to 4 subsequent simulation blocks).
Clothes Washer	470 W	3 W	45 - 60	Power varies dramatically between agitation and high-speed spin cycles; 470W represents a smoothed cycle average. ¹²
Clothes Dryer	2100 W	2 W	60 - 90	Generates high sensible heat gain. If vented outdoors, EnergyPlus must account for the exhaust air makeup drawing unconditioned air into the zone.
Television (LCD/LED)	100 W	1 W	Continuous w/ activity	Follows the exact duration of the "Leisure" activity block.

Desktop PC & Monitor	150 W	5 W	Continuous w/ activity	Follows the exact duration of "Work" or "Education".
Laptop / Tablet	45 W	1 W	Continuous w/ activity	Often runs on battery; charging load may be asynchronous to actual usage.
Hair Dryer / Grooming	1500 W	0 W	10 - 15	Captured under GSS 7 (Personal Care). Scale power by 0.33 for 30-min energy equivalence.
Refrigerator / Freezer	84 W (Avg)	N/A	Baseload	Do not assign to human activities. Assume a continuous sinusoidal or constant internal heat gain operating 24/7. ¹²

The Temporal Discretization Problem and Cycle Queues

A primary mathematical friction point in mapping 30-minute TUS data to physical appliances is the fundamental mismatch between the human activity duration and the machine cycle duration.

When a user's pre-calculated schedule flags a 30-minute block as "Household work" and the deterministic matrix triggers the dishwasher (930 W), the physical dishwasher cycle operates for 90 to 120 minutes. Therefore, the transformation algorithm must implement a rigorous

cycle queue. If an appliance with a characteristic duration $D_{app} > 30$ minutes is triggered at time t , the load algorithm must enforce a scheduled override that maintains the appliance's active power state for the subsequent $(D_{app}/30) - 1$ timesteps. This must occur regardless of what the human occupant's sequence dictates in those subsequent timesteps (e.g., the occupant may move from "Household work" to "Sleep", but the dishwasher continues to run). Failure to implement this queueing logic in the EnergyPlus schedule will result in artificially truncated load spikes and a severe underestimation of total daily volumetric energy consumption.

Conversely, for appliances with $D_{app} < 30$ minutes (such as microwaves, kettles, and hair dryers), applying the full nameplate rating across the entire 30-minute simulation block violates the conservation of energy, injecting massive, erroneous heat gains into the thermal zone. Therefore, the active power must be prorated across the 30-minute block. The injected schedule value for EnergyPlus must be calculated as:

$$P_{injected} = P_{rated} \times \left(\frac{D_{app}}{30} \right)$$

This deterministic scaling ensures that the integral area under the power curve (total Joules consumed) remains thermodynamically accurate for the building's internal heat gain and electrical grid load calculations.

Co-Presence Scaling: Sub-Linear and Linear Laws

When generating profiles for households comprising multiple occupants ($o > 1$), independent 30-minute activity streams frequently overlap. If Occupant A and Occupant B are both deterministically flagged as engaging in "Passive leisure" simultaneously, a naive addition of their individual load profiles will erroneously double the entertainment load. In reality, residential appliance utilization obeys strict behavioral physics, separating explicitly into "Shared" (Sub-Linear) devices and "Personal" (Linear) devices.

To resolve overlapping activity sequences without reverting to stochastic Markovian generation, the modeling framework must parse the aggregate household activity state through explicit mathematical scaling functions.

The Sub-Linear Law of Shared Utilization (Effective Occupancy)

For communal end-uses—most notably room lighting, television viewing, space-heating related auxiliary loads, and the initiation of large shared appliances like the oven or dishwasher—the addition of a second, third, or fourth occupant to the same activity yields rapidly diminishing marginal electricity demand. Two occupants watching a film do not power two primary televisions; two occupants eating dinner together do not require two electric ranges operating simultaneously.

The concept of representing this shared efficiency was rigorously formalized as "Effective

Occupancy" (EFF) by Richardson et al. (2009/2010) and is foundational to the CREST domestic demand models.¹ The effective occupancy represents the equivalent non-linear scaling multiplier for a room or dwelling when multiple occupants are active. Based on extensive statistical analysis of time-use micro-data and domestic electricity demand surveys, the scaling factor follows a piecewise sub-linear progression.

The recursive incremental formula utilized to calculate the marginal addition of the o -th occupant, derived from empirical household survey data, is explicitly defined as³:

$$EFF_o = (0.2545 \times o) + 0.6629 - EFF_{o-1}$$

When evaluating the aggregate effective occupancy mapping for N simultaneously co-present occupants engaged in a shared activity, the stabilized deterministic multipliers are fixed as follows¹⁶:

- $N = 0 \rightarrow EFF = 0.0$
- $N = 1 \rightarrow EFF = 1.00$
- $N = 2 \rightarrow EFF = 1.40$ (or conservatively **1.50** depending on specific lighting vs. HVAC boundary calibrations)
- $N = 3 \rightarrow EFF = 1.70$
- $N = 4 \rightarrow EFF = 1.90$
- $N \geq 5 \rightarrow EFF = 2.00$ (representing the saturation point of shared residential infrastructure)

Implementation Rule for Shared Devices: Whenever the sum of active occupants engaged in a shared activity (e.g., Passive Leisure, Eating, Socializing, Household Work) in timestep t equals N , the aggregate power demand injected into the EnergyPlus schedule for the triggered appliance is:

$$P_{total}(t) = P_{rated} \times W_{activity} \times EFF(N)$$

This deterministic scaling completely eliminates the need for Markovian "co-use" probability distributions. If the deterministic sequence dictates that three people are simultaneously watching TV, the calculated load is **$100 \text{ W} \times 1.7 = 170 \text{ W}$** . This elegantly represents the operation of one primary screen (100 W) alongside smaller secondary parasitic loads, room lighting expansions, or minor peripheral devices (70 W), accurately modeling the physical reality of a shared space.

The Linear Law of Personal Utilization

Conversely, specific end-uses scale linearly with the number of co-present occupants because they involve highly individualized equipment, isolated spatial computing, or direct volumetric consumption of resources.

Personal devices and end-uses include:

- Laptops and desktop computers (GSS 1: Work, GSS 8: Education)
- Hair dryers and grooming tools (GSS 7: Personal Care)
- Domestic hot water draw volumes (Showers/Bathing routines mapped to GSS 7)

Implementation Rule for Personal Devices: If N occupants are simultaneously engaged in a personal activity, the effective occupancy scaling rule is bypassed, and the relationship is

strictly linear ($EFF = N$). If three occupants are simultaneously flagged for "Work" (GSS 1) at 14:30 on a Wednesday, the computing load injected into the schedule is precisely

$3 \times 150 \text{ W} = 450 \text{ W}$. The model assumes distinct, non-shared infrastructure for these specific categorical classifications.

Chronological and Climatological Corrections: Seasonality

A well-documented analytical vulnerability of baseline Time-Use Surveys is their inherent temporal bias; longitudinal surveys are often conducted over limited weeks and fail to comprehensively capture behavioral shifts across deep seasonal divides. For a Canadian multi-year (2005–2030) EnergyPlus study targeting decarbonization, grid resilience, and building performance, failing to account for seasonality will result in severe underestimation of winter internal heat gains and overestimation of summer baseloads.²⁵

The Canadian Winter Intensification Effect

In high-latitude Canadian environments, the harsh winter season (broadly spanning December to March) forces a massive intensification of indoor residency times and radically alters the operational profile of residential appliances. This phenomenon, categorized in building physics as "winter intensification," drives up continuous active occupancy, shifts dietary and cooking habits toward hot, high-thermal-mass meals (dramatically increasing heavy oven usage over light microwave usage), and fundamentally alters spatial lighting demand due to severely reduced daylight hours.²⁵ As documented in Canadian system operator outlooks and stochastic load generating research, utility systems with strong seasonal variations in space heating demand also observe highly correlated behavioral plug-load and lighting spikes that traditional static schedules entirely miss.²⁷

Implementation of the Fischer synPRO Adjustments

To organically correct the flat TUS activity sequences without destroying the integrity of the user's pre-calculated deterministic 30-minute profiles, researchers strongly recommend

applying a macro-seasonal modulation factor across the baseline weights. The Fischer synPRO methodology, widely utilized for modeling robust European and high-latitude energy communities²⁹, expertly adjusts the probability mass and operational load factors of specific end-uses strictly based on the time of year.

For the Canadian deterministic framework, the seasonal correction should be applied dynamically to the mapping weights as a continuous mathematical function of the day of the year (d , where $1 \leq d \leq 365$).

1. The Lighting Irradiance Correction:

Lighting demand must never rely solely on the qualitative activity flag; it requires an active environmental irradiance override. In EnergyPlus, the final lighting schedule multiplier

$L_{mult}(t)$ should be formulated to incorporate the Effective Occupancy scalar alongside an inverse daylighting function:

$$L_{mult}(t) = W_{activity}(t) \times EFF(N) \times f(E_{ext})$$

Where $f(E_{ext})$ is an inverse function of the exterior global horizontal irradiance. If E_{ext} exceeds the spatial threshold required for comfortable indoor illumination (e.g., 200 W/m²), $f(E_{ext})$ approaches 0.0, smoothly suppressing the artificial lighting load. This dynamic interaction naturally enforces the massive winter lighting intensification inherent to Canadian winters—where daylight hours are truncated and exterior irradiance is low—without requiring structural alterations to the base TUS data sequences.

2. The Hot Water and Cooking Intensification Scalar:

For DHW pumping loads and primary Range/Oven usage (which are highly susceptible to seasonal behavioral shifts), a sinusoidal correction factor should be superimposed onto the baseline probability weights. To capture the depth of the Canadian seasonal trough and peak, the adjusted weight $W_{winter-adj}$ is calculated as:

$$W_{winter-adj} = W_{base} \times \left(1 + \alpha \cos \left(\frac{2\pi(d - 15)}{365} \right) \right)$$

Where $d = 15$ aligns the absolute peak of the cosine wave with mid-January, historically representing the deepest nadir of Canadian winter temperatures and behavioral indoor

isolation. Empirical distribution data suggests an amplitude factor of $\alpha = 0.15$ is appropriate for Canadian residential cooking and hot-water-related appliance loads. This effectively and deterministically increases the magnitude of these specific thermal loads by 15% in deep winter and symmetrically depresses them by 15% in mid-summer, flawlessly emulating the winter intensification effect within a completely rigid sequence.

High-Resolution Validation Frameworks for the

Canadian Context

No deterministic load generator can be confidently deployed into a predictive multi-year EnergyPlus framework without rigorous empirical validation. Because the overarching goal of such simulations is the accurate projection of electrical grid impacts, peak shaving potential, and nuanced building thermodynamics, the synthesized output must be meticulously compared against measured Canadian end-use datasets.

Archival Beausoleil-Morrison Datasets

The gold standard for high-resolution residential electricity validation within the Canadian context relies on two extensive, highly disaggregated field measurement campaigns conducted in Ottawa, Ontario, hosted by the Solar Energy and Building Physics Laboratory (SBE) at Carleton University.³²

1. Saldanha & Beausoleil-Morrison (2012)²: *Measured end-use electric load profiles for 12 Canadian houses at high temporal resolution* (Energy & Buildings 49, DOI: 10.1016/j.enbuild.2012.02.050). This dataset provides highly disaggregated, appliance-level circuit measurements crucial for validating the specific appliance fractional weights outlined in this report.
2. Johnson & Beausoleil-Morrison (2017)³²: *Electrical-end-use data from 23 houses sampled each minute for simulating micro-generation systems* (Applied Thermal Engineering 114, DOI: 10.1016/j.applthermaleng.2016.07.133). This significantly expands the sample size and provides excellent non-HVAC baseload mapping and whole-home profile shapes.

Data Acquisition and Format: Both of these pivotal datasets are publicly archived and readily accessible for BPS researchers via the Carleton University repository

(carleton.ca/sbes/publications/electric-demand-profiles-downloadable/).³² The empirical data is deliberately provided in two structural formats to facilitate different layers of research:

- *Logged Data:* These files contain raw, unprocessed 1-minute averages containing authentic measurement noise, periodic sensor dropouts, and missing records.
- *Processed Data:* A contiguous, mathematically filled, and smoothed sequence providing exactly one year of 1-minute interval data per house. For the purpose of validating the 30-minute deterministic EnergyPlus outputs detailed in this report, this *processed* dataset is essential. It must be block-averaged (integrated) up to 30-minute intervals to match the exact temporal discretization of the simulation output, allowing for direct comparative analysis.

Recommended Validation Metrics and Protocols

When validating the synthetically mapped appliance loads against the SBE empirical data, traditional statistical error metrics like Mean Absolute Error (MAE) or Root Mean Square Error (RMSE) are fundamentally insufficient and often deeply misleading. Because human activity is temporally volatile by nature, a spike in cooking loads at 17:30 in the model versus 18:00 in the empirical data will yield a massive RMSE penalty, despite the model being physically,

thermodynamically, and behaviorally accurate on a macroscopic level.

Instead, the validation protocol must eschew point-by-point tracking and focus on distribution density, aggregate physical dynamics, and geometric shape profiling:

1. **Load Shape and Peak Coincidence (Diversity Factor):** The primary validation step requires analyzing the aggregate behavior of 10 or more modeled homes against the average of the Johnson 23-house dataset.³⁶ The "Diversity Factor"—defined as the sum of individual maximum residential demands divided by the maximum demand of the aggregated system—must be computed. A mathematically successful deterministic appliance mapping will yield an aggregated load shape that smooths out individual household volatility, mirroring the distinct 17:00–20:00 Canadian evening peak without artificial stacking or simultaneous unrealistic synchronization.
2. **Energy Use Intensity (EUI) Distribution:** The validation algorithm must integrate the area under the synthesized power curves to calculate the annual total kWh/year for lighting, active appliances, and DHW separately. The probability density function of the modeled households' EUI must then be compared against the National Energy Use Database (NEUD) averages and the empirical sample means from the Saldanha 12-house set.² This ensures that while the timing may shift, the thermodynamic volume of energy injected into the building over a year is sound.
3. **Fréchet Distance and Dynamic Time Warping (DTW):** To compare the morphological similarity of the daily load profiles without harsh penalization for slight temporal phase shifts inherent to rigid 30-minute blocks, DTW should be employed. This algorithmic approach measures the geometric similarity between the synthesized sequence and the Ottawa empirical sequence by warping the time axis to align similar peaks. Demonstrating a high DTW similarity score proves that the deterministic mapping algorithm generates highly authentic Canadian residential load signatures, validating its deployment for macro-scale grid and building physics research.

Implementation Directives for EnergyPlus Environments

To finalize the integration of this deterministic appliance mapping framework into EnergyPlus, the numerical outputs and scaling laws must be structurally formatted to interact with the engine's solver. Rather than defining rigid, static Schedule:Compact objects that cannot dynamically respond to co-presence or seasonal shifts, the optimal methodology utilizes the EnergyPlus Runtime Language (Erl) via the Energy Management System (EMS).

The 30-minute predicted GSS activity codes (1 through 14) and the active occupant count (N) for the household should be passed into EnergyPlus as Schedule:File time series. The EMS program can then read these activity states at each specific timestep, execute the sub-linear

scaling functions (EFF) for shared devices, apply the fractional appliance mapping weights, initiate the cycle-queue logic for long-duration appliances like the dishwasher, run the Fischer seasonal winter-intensification cosine correction, and ultimately inject the final calculated Watts directly into OtherEquipment and Lights objects via EnergyManagementSystem:Actuator

commands.

This approach completely circumvents the limitations of stochastic Markov generation while transforming static time-use arrays into a highly dynamic, behaviorally authentic, and thermodynamically accurate simulation environment tailored specifically to the high-latitude intricacies and physical realities of the Canadian residential building stock.

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